

Subject agreement emerges from topic continuity, not topic shift: Corpus-based evidence from Hewramî, a West Iranian language

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Abstract

While the loss of subject agreement is often related to general processes of phonological attrition, the processes by which subject agreement emerges remain disputed. This paper tests one of the most influential approaches, the “topic shift” account of Givón (1976, 2017), which identifies a pragmatically-marked construction signaling topic discontinuity as the point from which subject agreement initially spreads. We investigate subject indexing in Hewramî, a West Iranian language, which exhibits differential subject indexing in past transitive constructions. Historically, West Iranian languages lost subject agreement in these constructions, but retained it elsewhere. Most contemporary languages have since restored subject agreement in past transitives, but in Hewramî such innovated subject indexing is not fully grammaticalized. We combine quantitative and qualitative methodologies on a sample of 2005 past transitive clauses from spoken Hewramî in order to identify the factors that influence the presence of an index. The quantitative analysis reveals that subject indexing is overall favoured in clauses that exhibit topic continuity, in particular those otherwise lacking an overt subject NP. In clauses with an overt subject NP, it is features associated with focus, rather than topic continuity, that inhibit indexing. We also find effects of person and animacy, with indexing most frequent with first and second person, and other human subjects. We find no evidence for the claim that a pragmatically marked construction for marking “topic shift” provided a model construction for these developments. Our findings constitute a novel and empirically well-grounded contribution to one of the most widely-researched domains of diachronic morphology.

Keywords: subject agreement, differential A indexing, topic shift, grammaticalisation

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I Introduction

There was a rich paradigm of subject indexing in all tenses in Old Iranian, but in Middle Iranian, it was lost in past transitive constructions, though maintained elsewhere. Subsequently, Iranian languages have undergone a range of different developments in their agreement systems in the past transitive domain. A particularly widespread development involved using clitic pronouns as agreement markers indexing past transitive subjects (A-past), but other outcomes are also attested. Taken together, the Iranian case study provides us with a remarkable window of opportunity for investigating the emergence of agreement in a sample of related languages. Generally, four different outcomes can be identified in contemporary West Iranian languages with regard to A-past indexing (see Figure 1 for the geographical distribution of these groups):

- I. Past transitive constructions still lack any subject indexing at all (e.g., Northern Kurdish, Zazaki)
- II. Clitic pronouns are deployed as indexes of A-past arguments, but they have retained (some of) their pronominal nature (e.g., Hewramî, some languages in the Tatic group)
- III. Clitic pronouns have been fully grammaticalized into agreement markers (e.g., Central Kurdish, Central Plateau)
- IV. Clitic pronouns were abandoned, and indexing via suffixal agreement morphology was restored in the past transitive clauses through extension from other contexts (e.g., Persian, Southern Kurdish)

In group III languages, clitic pronouns have become obligatory and are required in all past transitive clauses, regardless of pragmatic configuration. Essentially, they have become inflectional morphemes, thus they can co-occur with the controller in the same clause (see (1)). However, in terms of phonological bondedness (Norde 2009), they still show properties of clitics, rather than affixes. For example, they feature freedom of host selection: in (1a) the clitic attaches to the direct object; in (1b) it attaches to the derivational preverb. For Central Kurdish, the placement rule for past transitive subject indexing is that the clitic attaches to the first stress-bearing item in the verb phrase (VP), hence subjects are not viable hosts (see Öpengin 2019 for details). With minor variations, a similar rule holds for Hewramî.

(1) Central Kurdish (MacKenzie 1962)

- a. *duxtôr hālan ôtombîl=î girt*
 doctor immediately car=3SG:A¹ take.PST
 'The doctor immediately took the car.'
- b. *qāhîça haṭ=î-bird*
 carpet PVB=3SG:A-carry.PST
 'The carpet carried him up.'

¹ The grammatical relations expressed by each bound person/number formative are represented in the glossing using capital letters following the relevant person forms. See the list of all abbreviations at the end of the paper.

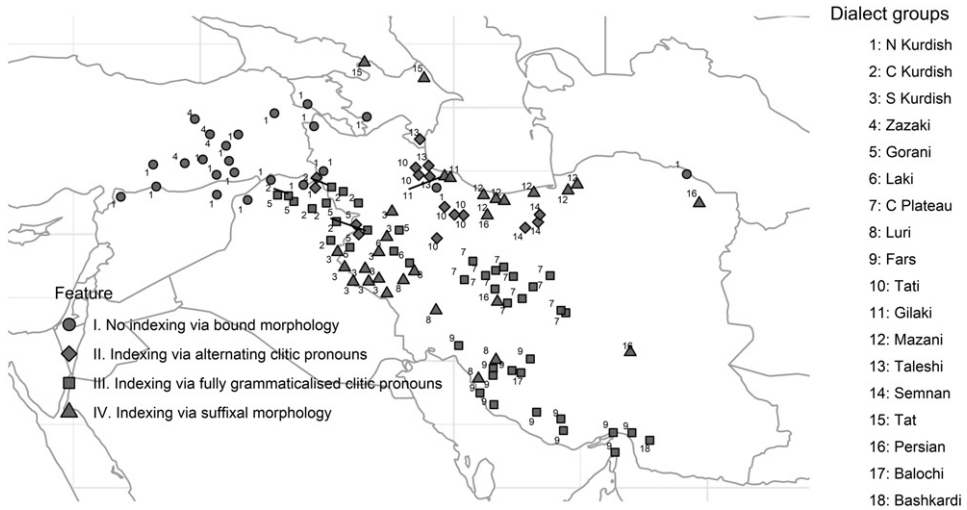


Figure 1: Indexing A-past arguments across Western Iranian – a sample map

A related feature of clitic pronouns in Iranian languages is that contrary to person-indexing suffixes in these languages, they are not prosodically integrated into the host element. These properties indicate that the morphological status of formatives as clitics or affixes is not (necessarily) a good diagnostic for distinguishing agreement from pronominal anaphora, confirming Kibrik's (2011: 185) point that a high degree of morphological and prosodic boundedness does not reliably correlate with obligatoriness of the marker concerned (see also Siewierska 2004b: 163 for languages where affixal indexes behave similarly to pronouns in that they are in complementary distribution with coreferential nominal arguments). In fact, existing scholarship on languages like Central Kurdish, illustrated in (1a) and (1b), converges on the analysis of the clitic pronouns as agreement morphology, despite the mobile clitic nature of the formatives concerned (see Samvelian 2007; Haig 2008; Öpengin 2013; among many others).

Group II languages deploy clitic pronouns as subject indexes, but unlike Group III, not every past transitive clause contains an index. There are various factors that inhibit the presence of an index. One possibility is for the index to be in complementary distribution with a co-referent subject NP in the same syntactic domain. In (2a), *varg-e* is the A argument, and cannot be doubled by a pronominal index. In (2b), the 3SG clitic *=eş* resumes the absent A argument.

(2) Chali (Tati group; Mohammadirad 2020: 390)

- a. *varg-e jeftak be-zandi=o*
 wolf-OBL leap PFV-hit.PST=and
 'The wolf leapt

- b. *şekar=eş paşindi*
 prey=3SG:A throw.PST
 and took down the prey.'

There is considerable terminological disparity in the way systems such as Chali are analysed. In terms of Siewierska's (2004b) three-way typology of person indices, the system would be considered "pronominal" (the clitic pronoun is in complementary distribution with its NP controller). Haspelmath (2013: 208) refers to elements such as $=e\varsigma$ in (2b) as "pro-indexes", while Kibrik (2011: 197) refers to them as "bound alternating pronouns", noting that they are typologically rare. Here we adopt the terminology of Kibrik (2011) in referring to this kind of complementary distribution as "alternating", thus the clitic pronoun in (2b) is an alternating index. Alternating indexes have been attested in other Iranian languages, such as varieties of Tati, Taleshi, and Semnani (Mohammadirad 2020: 159–162). The same feature occurs, seemingly to a limited extent, in Northern Kurdish dialects located in the transitional zone with Central Kurdish (Öpengin & Mohammadirad 2022). The Hewramî dialect of Tekht under investigation here is also treated as Group II, but it is not an alternating system like Chali. Rather, a bundle of pragmatic and semantic factors determines the presence or absence of the index, which we investigate below. We follow Just (2024) in referring to this kind of system as "Differential A Indexing", to avoid potential confusion with "DAI" used in the broader sense of "Differential Argument Indexing". In fact it is possible to interpret the alternating system of pronoun and co-nominal NP illustrated in (2) as a sub-type of Differential A Indexing, because the presence or absence of the conominal is primarily a consequence of pragmatic context (i.e., reflects speakers' choices), and in this sense is comparable to, for example, absence or presence of an index triggered by topicality of the referent (see Haig 2018b for this position). This is more a terminological than a substantive issue; for the time being we adopt the more traditional view which would distinguish alternating A indexing from Differential A Indexing.

The array of different subject indexing systems documented in the West Iranian languages, coupled with historical data going back more than two millennia to Old Iranian, make West Iranian one of the most fruitful arenas for investigating the emergence of subject agreement. In this paper, we focus on Hewramî, a Group II language that shows a partially, but not fully, grammaticalized system of subject agreement. There is relatively little detailed research on these systems and the nature of the conditioning factors that determine the presence of the index. But these systems are crucial in that they provide evidence for the intermediate stages in one of the most widely-researched topics in diachronic morphosyntax, namely how an inflectional paradigm of agreement markers grammaticalizes from erstwhile pronominal elements.

Before proceeding, some terminological conventions are in order. We use the following abbreviations to refer to the core arguments within the clause: S: intransitive subject; A: transitive subject; O: transitive object. The term "flagging" is reserved for overt marking of predicate-argument relations that is spelled out on the argument itself. Following Haspelmath (2013), we use the term "indexing" as a cover term for both "grammatical agreement" (e.g., *Mary walk-s in the park*) and "anaphoric agreement" (e.g., *John_i went out. He_i bought a book*). The term "agreement" is reserved for cases where the index obligatorily reflects an argument regardless of the presence or the absence of the coreferent NP in the clause (e.g., Italian: *(lui) vien-e* [3SG come.PRS-3SG] 'He comes'), and regardless of other contextual factors. Agreement in this sense parallels

what Haig (2018b) refers to as “obligatory” indexing, while Haspelmath (2013) refers to “cross indexing” and Kibrik (2011) to “tenacious bound pronouns”.

One diagnostic for recognizing different degrees of agreement systems is the extent to which clauses are permitted with no overt segmental expression of the subject argument at all – neither independent pronoun or NP, nor index. We consider this a potentially informative diagnostic for assessing the degree to which pronominal elements have grammaticalized towards agreement markers.² The rationale is as follows: it is cross-linguistically typical for pronouns to be dropped in certain syntactic configurations, for example, in same-subject sequences of tightly knit temporal events, where even English prefers zero anaphora (see Kibrik 2011: 177–178 and Schiborr 2023, among others, on same-subject clause sequences as cross-linguistically preferred contexts for zero anaphora).³ If a language retains its pronominal index in such contexts, we take this as a significant shift in the functionality of the relevant item, from pronoun towards obligatory agreement marker. But in order to detect the extent of such usage, as well as identifying larger contextual factors triggering the presence or absence of the index, larger stretches of connected discourse need to be carefully analysed. This paper is the first attempt to conduct such an analysis on an Iranian language that exhibits a variable system of indexing.

The paper is structured as follows. §2 provides an overview of approaches to the rise of subject agreement. §3 contains a thumbnail sketch of Hewramî, the corpus, and the research questions. §4 introduces the main features of A-past indexing in Hewramî. §5 presents the multivariate approach to Differential A Indexing in Hewramî and lays out the relevant predictors. §6 gives a detailed account of subject focus as the strongest factor triggering Differential A Indexing. §7 summarizes animacy and clause type as factors with weak effect on Differential A Indexing, while §7 investigates the link between indexing and topic shift. §8 summarizes the main results and considers the broader implications for diachronic morphology.

2 Approaches to the grammaticalization of subject agreement

The grammaticalization of subject agreement from former pronominal elements is widely attested and has been intensely researched over the last 50 years. Broadly speaking, there are three usage-based approaches (Siewierska 2004b: 263–268): the topic dislocation analysis (Givón 1976, 2017),⁴ the Accessibility Theory (AT, Ariel 2000); and

² This suggestion is already prefigured in Himmelmann (1999), a corpus-based study on (the lack of) zero anaphora with “Undergoer-Oriented Predicates” in Tagalog narrative texts. This study came to our attention too late for inclusion in the main text, but the parallels between the findings and our own are unmistakable.

³ Kibrik’s (2011: 178) apparent counter-examples to pronoun deletion in English all involve pronouns in combination with an auxiliary (e.g., *she’s*). Pronouns in this context are rarely dropped without also dropping the attached auxiliary, which impacts overall rates of omission (see discussion in Wagner 2018). This is further support for the impact of language-specific structural factors in determining zeroes, rather than solely functional and communicative factors (see Haig & Schnell under review).

⁴ Siewierska (2004b) and Ariel (2000) use the term “NP detachment”, but Givón (1976) originally refers to “dislocated topics”, and generally to “topic shift” and “topic dislocation”.

frequency-driven morphologization (FDM) (see Bybee 2006, among others). Given the sheer volume of available research, here we consider primarily the topic dislocation analysis, as the most well-known and influential theory for the grammaticalization of subject agreement. We also discuss AT, and so-called complementarity effects in subject indexing, according to which the presence of an index and the presence of an overt subject constituent are mutually incompatible, or at least show significant interactions in quantitative corpus-based analyses.

The topic dislocation analysis, as originally developed in Givón (1976) and more recently summarized in Givón (2017), states that agreement arises from anaphoric pronominalization in topic-shift constructions. Thus, the origins of agreement are linked to a pragmatically marked construction, restricted to contexts where a topic shift is intended ('as for X, ...'). This construction then extends into contexts where a topic-shift is no longer implied, and speakers "re-analyze it as the neutral syntax" (Givón 1976: 154), so that it becomes the norm. Givón (1976: 155) illustrates the assumed re-analysis with the Pseudo-English example in (3).

- (3) a. *The man, he came.* (Topic dislocation construction, "topic shift")
 b. *The man he-came.* (Pragmatically neutral subject-verb construction)

Givón (2017: 63) cites a number of studies in support of the claim that left-dislocation is a widely-used strategy for re-activating referents that are "beyond the currently active clause-chain". Functionally, "L-dislocation (or NP fronting) is used systematically as a chain/paragraph-initial device, signalling both **referent re-activation** and the opening of a **new thematic unit**" [emphasis in original]. The central role assigned to this construction in the emergence of subject agreement is justified as follows (Givón 2017: 81–82):

The reason why L-dislocation is of great interest for the evolution of pronominal agreement is because – at least *prima facie* – it displays two features that can transform into obligatory pronominal agreement, when the paratactic L-dislocation construction is condensed into a simple syntactic clause:

- a full NP antecedent co-referent to the following anaphoric pronoun; and
- an unstressed anaphoric pronoun adjacent to the verb and ready to cliticize.

For Givón, the change from dislocated topic construction to subject agreement is a natural outcome of what he considers to be a more general diachronic tendency, namely the "condensation" of a paratactic structure such as (3a) into a "syntactic" structure (3b), or more generally, the emergence of morpho-syntax from discourse-pragmatics ("syntacticization", Givón 2017: 102). In the case of the topic shift construction, this tendency yields "as by products obligatory subject or object agreement" (Givón 2017: 82). While a number of languages possess agreement constructions that could be considered outcomes of such a process, there is a paucity of evidence plausibly documenting the individual stages of the assumed development. Givón (1976) cites data from pidgin and creoles, where erstwhile personal pronouns occur as agreement clitics, but

these data illustrate an outcome, rather than the process itself. Forty years after the original claims, Givón's (2017: 87) more recent appraisal is that while the original claims are "plausible", the "potential role of L- or R-dislocation in the diachronic rise of pronominal agreement remains to be fully assessed", and "much of the detail remains to be worked out" (Givón 2017: 99).

Other scholars have raised issues with the dislocated topic account. Siewierska (2004a) and Ariel (2000) point out that dislocated topics would most frequently be third person topics, such as 'the man' in (3a), while first and second person subjects would be infrequent as dislocated topics. Yet subject agreement appears to emerge first for first and second person (see below). The primacy of first and second person in agreement systems is better handled in AT, which relates the rise of agreement to the accessibility of referents in discourse. Accessibility is considered to reflect the degree to which a particular referent is identifiable to the listener at a given point in the discourse. Accessibility is a product of "antecedent salience", which includes features such as person, animacy, topicality of the antecedent, and "unity", which is primarily concerned with distance between the referring form and its antecedent.

Ariel (2000) claims that the highest accessibility forms will grammaticalize earliest into agreement markers. Crucially, she considers first and second person pronouns to be higher in salience than third person pronouns (Ariel 2000: 211). The AT account thus predicts that first and second person pronouns will grammaticalize earliest into agreement markers. Given the widespread consensus that it is first and second person forms that grammaticalize earliest into agreement markers (van Gelderen 2011; Siewierska 2004b), the AT account would seem preferable to topic dislocation. However, although Ariel (2000) repeatedly stresses that first and second person forms are the highest in accessibility, thus most likely to grammaticalize first into agreement markers, it is less clear how high accessibility in itself should lead to the grammaticalization of an anaphoric pronoun to an agreement marker.

A further drawback of the topic-dislocation account is the paucity of convincing evidence for the pivotal role of an initially pragmatically marked structure. It is well-known that subjects are generally encoded by reduced referential devices (zero, pronouns, bound indexing, Du Bois 1987; Ariel 2000), so we entertain doubts on the impact that pragmatically-marked and infrequent topic-shift constructions may have had (see below for figures from the Hewramî corpus). More generally, we fail to see any inevitability in the process of pragmatically-marked constructions losing their pragmatic force and becoming the neutral or default construction ("depragmatization"). For example, passive constructions may be linked to specific information-structural configurations but are not necessarily destined to be reanalysed as pragmatically neutral active constructions. Rather, they may simply remain as an alternative means of perspectivizing an event, or go out of use, or undergo other diachronic changes.

Alternative accounts downplay the importance of some pragmatically-marked construction as the model for the expansion of agreement, considering more general structural configurations as the prerequisite for change. Fuß (2005), for example, posits a primarily structural motivation in the emergence of subject agreement in shoring up gaps in defective paradigms; this would obviously apply to the Iranian case, where subject

agreement was entirely lost in the context of past transitive clauses. Translated into a usage-based approach, this would suggest that subject agreement is most likely to re-assert itself in those contexts where otherwise a subject constituent is absent (i.e., would suggest a degree of complementarity between overt subject NP and index): “if agreement on verbs helps identify the referents of arguments, then agreement should be favoured where arguments are likely to be non-overt (and vice versa)” (Nichols 2018: 857). Nichols (2018) refers to this as the complementarity principle. In a corpus-based approach, complementarity is a gradient metric: if there is a statistically significant association between, for example, the presence of a pronoun and the absence of an agreement marker, then this is considered as support for some degree of mutual influence.

The available research on this complementarity in discourse yields somewhat mixed results. Givón (2017: 85–86) presents a small case study based on written Modern Hebrew fiction. In Hebrew, some verb forms (perfective and irrealis) have rich paradigms of affixal subject agreement, distinguishing person, number, and some forms also distinguish gender, while imperfective and participial verb forms have defective paradigms distinguishing only number and gender. If complementarity were operative, we could expect a higher frequency of zero subjects with verb forms carrying rich agreement than with verb forms with weaker agreement. This prediction is borne out in Givón’s sample: 16% zero subjects with weak agreeing verbs ($N = 206$); 74% zero subjects with rich agreement ($N = 334$). However, Nichols’ (2018) study of Ingush discourse reports no significant increase of indexing in the absence of a subject constituent but notes that indexing in this language is based on gender, not on person. From a broader comparative perspective, Berdicevskis et al. (2020) provide evidence for a trade-off between agreement morphology and overt pronouns, based on Slavic data and a larger typological sample, though they note that this trade-off is most evident in Eurasia. Below, we consider complementarity effects in the Hewramî data, but we do not take complementarity in the sense of strict mutual incompatibility, but rather in the sense of quantitative trade-off effects – for example, a statistically significant quantitative effect of the presence of agreement on pronoun expression, or vice versa.

Returning to Iranian, one line of relevant scholarship has followed Bynon (1979) in adopting a Givón-style explanation for the rise of subject agreement from erstwhile clitic pronouns. Jügel (2009) and Jügel & Samvelian (2020) suggest that obligatory A indexing in Central Kurdish, a Group III Iranian language illustrated in examples (1a) and (1b) above, would have developed from a “hanging topic” construction, attested in Middle Persian (see discussion below). Haig (2018a), on the other hand, considers that the presence of clitic doubling is not the crucial diagnostic for recognizing the transition from anaphoric pronoun to agreement marker. Rather, he points to the existence of a preceding phase characterized by increased discourse frequency of pronouns in contexts which otherwise prefer zero. These include same-subject clause sequences (*Mary came in and Ø sat down*), which cross-linguistically strongly prefer zero (Schiborr 2023). The intrusion of pronouns into such contexts would be in principle independent of the kind of pragmatically-marked constructions that are claimed to act as catalysts in the topic dislocation scenario.

In sum, all research converges on assigning a pivotal role to some notion of discourse topicality, either defined loosely in terms of high “accessibility”, or more narrowly in terms of “maximal referential continuity” (Givón 2017: 100), that is, reference to immediately preceding and persistent entities at a given point in discourse. For Givón (2017), this means that subject agreement is related to “topic agreement”. Somewhat paradoxically, however, Givón’s topic dislocation approach proposes that the catalyst for the development of subject agreement is a pragmatically marked construction type deployed to signal a *disruption* of expected topic continuity. While this account has the advantage of identifying a structure that displays the structural hallmarks of syntactic agreement, it has proven difficult to identify the missing links between topic dislocation constructions and fully obligatory syntactic agreement, and the main motivation for assuming such an extension is a fairly vague appeal to “depragmatization” and “condensation” of paratactic structures. Alternative accounts remain vague on the actual pathways of change, but appeal more generally either to abstract structural mechanisms such as shoring up defective paradigms (Fuß 2005), or a concept of “accessibility” which subsumes, but is not fully equal to, the concept of topicality in the sense of topic continuity. We also considered complementarity effects, which assume that indexing would be more frequent in contexts otherwise lacking any exponent of the subject referent, and vice versa. If such effects are documented in contemporary usage, it is not unreasonable to assume that they will have been operative in the diachronic emergence of subject agreement.

3 Hewramî typology, data sources, and research questions

Hewramî (Glottocode: hawr1243) is a member of the Gorani subbranch of the Iranian languages. The classification according to Glottolog is: Indo-European: Iranian: Central Iranian: Northwestern Iranian: (Adharic:) Gorani: Hewramî. It is spoken in a high mountainous region at the heart of the Kurdish-speaking region along the western border of Iran and neighbouring areas in Iraqi Kurdistan. Hewramî dialects are traditionally divided into three major groupings: Tekht, Luhon, and Jawero. Geographically speaking, these varieties are spoken in the centre, south, and east of the greater Hawraman region, respectively. MacKenzie (1966) is a sketch grammar of the Luhon variety. Mohammadirad (2026) provides a comprehensive grammatical description of the Tekht variety.

3.1 Basic facts about Hewramî grammar

Hewramî has a basic AOV word order. Like most Iranian languages, it has retained split ergativity, meaning that the alignment pattern is nominative-accusative in clauses with present tense verbs, and ergative-absolutive in clauses with past tense verbs. Ergative alignment is most consistently evident in the pattern of argument indexing: past transitive clauses exhibit a specific paradigm of subject indexing (A–past), which differs from subject indexing in intransitive clauses (S), and from object indexing in the past tense (O–past), as shown in Table 1. This paradigm results from the merger of

Table 1. Bound argument indexing in Hewramî (Mohammadirad 2026)

	S/A-present	S/O-past	A-past
1SG	-û	-anê	=im
2SG	-î	-î	=it
3SG	-o	-Ø (m); -e (f)	=is
1PL	-mê	-îmê	=ma
2PL	-dê	-îdê	=ta
3PL	-a	-ê	=ša

genitive/dative and accusative sets of clitic pronouns from Old Iranian languages. The same clitics are also used as markers of adnominal possession and as complements of adpositions, where they are in strict complementary distribution with free pronouns or coreferential NPs. In A-past function, they remain clitics, rather than affixes, in that they exhibit freedom of host selection. Example (4) illustrates the ergative pattern of indexing.

(4) Hewramî

a. *wit-ê*

sleep.PST-3PL:S
'They slept.'

b. *berd-ê=ša*

take.PST-3PL:O=3PL:A
'They took them.'

Example (4a) illustrates S-indexing in a past tense intransitive clause. The verb obligatorily takes the relevant suffix (3PL from the S/O paradigm), which is always required, regardless of the presence or absence of an overt subject NP, and is always a suffix in this slot on the verb.⁵ Example (4b) illustrates 3PL indexing of a past transitive subject (the A-past paradigm). The index itself is a clitic pronoun, which will generally attach to the initial element of the VP (here the verb itself; see (1a)–(1b) for other host possibilities). Note that the clitic index follows other agreement morphology, for example, the suffix *-ê* indicating plural number of the direct object. Note further that these clitic pronouns are homophonous with (and etymologically identical to) the set of clitics expressing pronominal possessors. Finally, the clitic index of A in (4b), unlike the suffixal index for S in (2a), is not fully obligatory; it can be omitted under circumstances

⁵ See Mohammadirad (2025a) for a corpus-based analysis of object agreement in Hewramî.

which will be explored in the remainder of the paper. Example (5) contrasts absence (5a) and presence (5b) of the A-past index.

- (5) a. *pađša-y* *fermawa*
 king-M.SG.OBL say.PST
 ‘The king said.’ [ME.29]
- b. *pađša-y* *fermawa=š*
 king-M.SG.OBL say.PST=3SG:A
 ‘The king said.’ [ED.199]

Hewramî has a two-term case system: ‘direct’ and ‘oblique’. Direct and oblique are terms traditionally used in Iranian linguistics and in the literature on two-term case systems (Arkadiev 2008), roughly equivalent to ‘nominative’ and ‘non-nominative’, respectively. Ergativity is manifested by oblique case marking of those transitive subjects which are nouns and third person pronouns, though it may be absent under certain conditions (see §5.1.2 below). Table 2 exhibits nominal flagging on nouns.

Personal pronouns have retained case marking only in the third person (see Table 3). Speech act pronouns have lost case distinctions, but the existing forms are derived from the historical oblique case.

Table 4 lays out the distribution of direct and oblique cases in flagging core arguments of the verb, applicable to third person pronouns and nouns.

The following examples show the ergative alignment of Hewramî through the noun *jenî* ‘woman, wife’. In (6a), the intransitive subject is in the direct case, with a clitic possessive pronoun attached; note the homophony between 1SG possessive clitic in (6a) and 1SG A-past index in (6b). In (6b), *jenî* (direct case) is the O of a transitive verb, and the A-past index attaches to it, because it is the first item of the VP, in accordance with the general rule for clitic placement. In (6c), ‘woman’ takes the F.SG form of the oblique case because it is the A of a past transitive clause, while the O ‘kohl pot’ remains in the direct case, and predictably hosts the 3SG clitic index for the A-past.

- (6) a. *jenî=m* *merdê=ne*
 wife.F.SG.DIR=1SG:POS die.PST.PTCP.F=COP.3SG.F:S
 ‘My wife (S) died.’ [ÇK.26]
- b. *jenî=m* *ardê=ne*
 wife.F.SG.DIR=1SG:A take.PST.PTCP.F=COP.3SG.F:O
 ‘I took a wife (O).’ [JM.20]
- c. *jenê* *kilêtan=iš* *nîya-Ø* *baxete=š*
 woman.OBL.SG:F kohl_pot.M.SG.DIR=3SG:A put.PST-3SG.M:O arms=3SG:POS
 ‘The woman put the kohl pot in her arms.’ [DB.233]

3.2 The Hewramî corpus

While the basic facts of Hewramî alignment, as just sketched, are comparatively well understood, little is known regarding the impact of discourse-related factors. In order

Table 2. Inflection of nouns in Hewramî

	DIR	OBL
SG.M	-Ø	-î
SG.F	-Ø	-ê
PL	-ê	-a

Table 3. Case inflection on pronouns

	DIR	OBL
1SG	<i>min</i>	<i>min</i>
2SG	<i>to</i>	<i>to</i>
3SG.M	<i>ađ</i>	<i>ađî</i>
3SG.F	<i>ađe</i>	<i>ađe</i>
1PL	<i>ême</i>	<i>ême</i>
2PL	<i>(ê)şme</i>	<i>(ê)şme</i>
3PL	<i>ađe</i>	<i>ađîşa</i>

Table 4. Case marking of core arguments

	Direct case	Oblique case
TAM based on present stem verbs	S, A	O
TAM based on past stem verbs	S, O	A (conditioned)

to investigate these aspects, two purpose-built corpora of naturalistic spoken Hewramî have been compiled and annotated. The first corpus is a collection of texts containing 15 narratives, totaling approximately 10,000 words (Mohammadirad 2025b). The second corpus is a collection of oral folktales recorded by Masoud Mohammadirad in Hawraman in August 2023. The corpus features 31 narratives and consists of approximately 25,000 words (Mohammadirad in prep.). Overall, seven narrators (including two women and five men, all over 50) were recorded telling folktales, local anecdotes, and life stories.

The two corpora were annotated for a range of semantic and pragmatic variables that are presented and discussed in §§5–7. These include animacy, constituent order, part of speech of A, and clause type. The dependent variable is the presence or absence of A indexing. We annotated all past transitive clauses, a total of 2,005 across both corpora, for the presence or absence of A indexing. Past transitive clauses with zero subjects make up 80% (1,595 out of 2,005) of the sample, and as seen later, feature 98% of A indexing. As clauses with zero subjects feature near grammaticalization of the A index and almost no variation, we only tagged clauses with overt A arguments (410 in total), which show 63% indexing across the full set of predictor variables presented in the following sections.

The corpus data were initially annotated in Excel. The annotated corpus is publicly available in a CSV file in the Zenodo repository (Mohammadirad & Haig 2025). The subsequent investigation of indexing in Hewramî is based on the statistical analysis of these corpora.

3.3 Research questions

In Hewramî, we find the comparatively unusual case of an Iranian language with partial, or differential, A indexing, as opposed to the obligatory indexing found in most Iranian languages. Given that the Hewramî system involves morphemes cognate to the obligatory indexes of related languages, it is a plausible proxy for reconstructing the earlier stages of the emergence of agreement prior to full grammaticalization. In the remainder of the paper, we analyze the Hewramî data in order to address the following research questions:

- i. What factors condition the presence of subject agreement in contemporary Hewramî?
- ii. To what extent is the presence of an index influenced by the presence of a coreferent A-past NP (complementarity hypothesis)?
- iii. What, if any, is the relationship between topic shift of the A-past NP and the rise of indexing?

4 A-past indexing in Hewramî: accessibility effects

As seen in Table 1, a paradigm of clitic pronouns indexes A-past arguments. These pronominal clitics are generally present in past transitive clauses, but under conditions to be discussed, may be absent. The first conjunct of (7) exhibits the presence of both an index and an overt subject constituent, while in the second part (with a subject change), an index is lacking. Example (8a) likewise shows the absence of an index in the presence of an overt nominal subject, while (8b) does contain an index, but no overt nominal subject.

- (7) *min řenc=im da=w tò berd*
 1SG toil=1SG:A give.PST=and 2SG take.PST
 ‘I toiled, and you took (the credit)!’ [YX.16]

- (8) a. *řosem-î nast hur-wéz-o ta da*
 PN-M.SG.OBL NEG.let.PST PVB-rise.PRS.SBJV-3SG:S so PTCL
bi-zan-o çêş ker-o
 SBJV-know.PRS-3SG:A what do.PRS-3SG:A
 ‘Rosam did not let him (Bezhan) rise (from the ground) so that he (Bezhan) could figure what to do.’ [BM. 149]

- b. *aza řemşêr=iş kêşt-Ø*
 quickly sword=3SG:A pull.PST-3SG:O
 ‘Quickly, he (Rosam) pulled out (his) sword.’ [BM. 150]

Table 5. Rate of indexing of overt A-past NPs vs. zero subjects

	N past tr. clauses	A is indexed		A is not indexed	
		N	%	N	%
Zero subject	1,595	1,566	98	29	2
Overt subject	410	257	63	153	37
Total corpus	2,005	1,823	91	182	9

These examples suffice to illustrate that there is not strict complementarity of overt subject and index in Hewramî: both may co-occur in the same clause, as in (7). Nevertheless, when the full range of data is taken into consideration, it becomes evident that indexing is overall more likely in the absence of an overt subject constituent, with (8b) being a typical example. Table 5 provides the overall frequency of indexing among all the past transitive clauses included in the sample.

It is evident that the overwhelming majority of transitive clauses derived from the past stem verb *do* in fact contain an index for the A (91%, 1,823 out of 2,005), as indicated in Table 5. These figures suggest that, at least in terms of frequency, the A index is at an advanced stage of grammaticalization towards an agreement marker. Table 5 also shows that in clauses lacking an overt subject NP, indexing is almost categorically required (98%). In other words, what is strongly dispreferred are clauses containing no overt exponent of the subject argument whatsoever. Thus, where an overt subject NP is absent (as in a large majority of transitive clauses, confirming well-established typological patterns, see Du Bois 1987; Haig & Schnell 2016; among many others), a pronominal index must be present. This kind of pattern does not comply with a standard definition of complementarity, which rules out co-occurrence of overt NP and index in the same clause (this is actually quite common among the set of clauses with overt subject NP, see Table 6), but it can be considered a weak complementarity effect in that absence of an overt subject NP strongly predicts presence of an index, but not vice versa. The first finding, then, is that in Hewramî, subject indexing is near-categorical in clauses that otherwise lack an overt subject constituent. Such clauses are typically found in chains of same-subject clauses, such as (9). The subject ‘the child’ is introduced via a lexical NP in the first clause, and resumed in the subsequent clauses through the clitic index *=iʃ*.

- (9) a. *zawle-(e)ke* *ama-Ø* *luwa-Ø* *biz(e)-û*
 child-DEF.DIR.SG.M come.PST-3SG:S go.PST-3SG:S goat.F-EZ.GEN
mama-lê-(ê)we *ʃot=iʃ* *ward-Ø*
 grandmother.F-DIM.F-INDF milk.M=3SG:A eat.PST-3SG.M:O
 ‘The child_i went and Ø_i drank milk from a grandmother’s goat.’ [KŞ.25]
- b. *duwê* *dehf(e)-ê* *ward-Ø=iʃ*
 two time.F-PL.DIR eat.PST-3SG.M:O=3SG:A
 ‘He drank milk [from the goat] two times.’ [KŞ.26]

Table 6. Indexing A-past arguments according to the referent of the A-past role

PoS of A	N clauses	A is indexed		A is not indexed	
		N	%	N	%
Zero subject	1,595	1,566	98	29	2
SAP pronouns	70	55	79	15	21
Third person pronouns	33	21	64	12	36
Nouns	270	162	60	108	40
Indefinite pronouns	30	17	57	13	43
Wh-pronoun	7	2	29	5	71
Total overt As	410	257	63	153	37

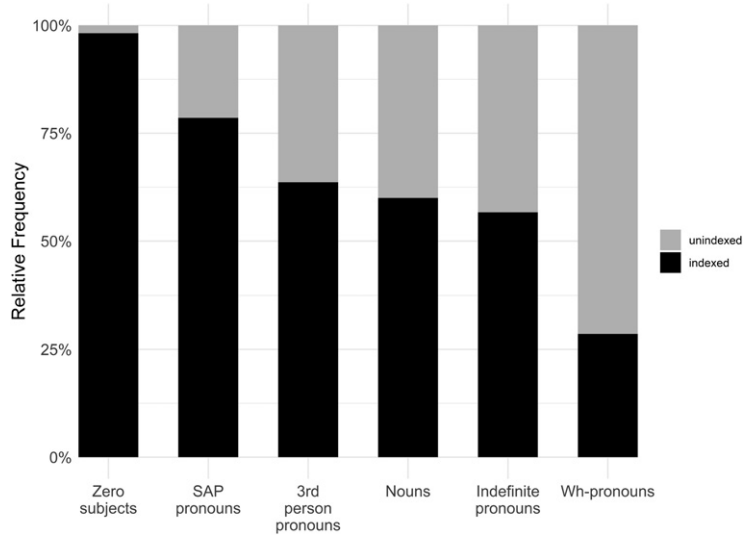


Figure 2: A-past indexing in Hewramî and the lexical category of the coreferent NP

The use of the pronominal index in this kind of construction is quite different to what would be expected of an independent anaphoric pronoun. In all Iranian languages, past and present, the norm in clauses like (9a) is zero, that is, dropping the subject pronoun. The presence of the clitic pronoun in the second conjunct thus already indicates that these clitic pronouns are no longer behaving in a manner typical of pronouns.

Turning now to those clauses that do contain an overt subject constituent, 63% also contain a pronominal index. Table 6 provides a breakdown of these cases, differentiated for person value, pronoun versus NP, and different kinds of pronoun, while Figure 2 visualizes the results.

As can be seen, the rate of indexing varies considerably according to the nature of the overt subject NP: SAP pronouns have the highest rate of A-past indexing, followed by

Table 7. Pairwise testing for significance of values in Table 6

Categories	Fisher exact value	p < 0.05
Zero vs. SAP	< 0.00001	yes
SAP vs. 3P pronouns	0.1492	no
SAP and 3P pronouns vs. all nouns (proper and common)	0.0158	yes
SAP vs. all nouns	0.005	yes
SAP vs. indefinite and wh-pronouns	0.0076	yes
All nouns vs. indefinite and wh-pronouns	0.375	no
Indefinite pronouns vs. wh-pronoun	0.232	no

third person pronouns, nouns, indefinite pronouns, and the wh-pronoun ‘who’. When tested pairwise, not all differences reach significance, but this is partly a consequence of low absolute numbers in some of cells. A selection of combinations is provided in Table 7.

To summarize, we have seen that the best predictor for indexing so far is the complete absence of an overt A argument in the clause (zero subject): 98% of such clauses contain an index. However, the converse does not hold: presence of an overt subject does not reliably predict absence of an index. Among clauses with an overt subject, a significant majority (more than 60%) still include an index. Among these, we identify increasing rates of indexing approximately correlating with increasing rates of accessibility, as defined by Ariel (1988, 2000): highest rates of indexing are associated with first and second person pronouns, followed by third person pronouns. This aligns with Ariel’s (2000) claim regarding the maximum accessibility of first and second person pronouns, but also with studies that point to pronouns generally as the most likely triggers for agreement in Differential Subject Indexing systems (e.g., Bentley 2018: 1291 on VS-constructions in Italo-Romance dialects). The lowest rates of indexing are associated with indefinite and interrogative pronouns, which by definition do not code accessible referents. This is suggestive of an inhibitive role of focality on indexing: subjects that are focal (see §6), for example interrogative pronouns, are the least likely to trigger indexing. Provisionally, we can summarize in (10) the findings in the form of the hierarchy of likelihood of triggering an index.

- (10) Hierarchy of likelihood of subject indexing in Hewramî
 zero subjects > SAP pronouns > third person pronouns > nouns (proper and common) > indefinite pronouns > wh-pronoun ‘who’

In §5, we build on this analysis and consider additional factors and their respective contributions by conducting a multivariate analysis using conditional inference trees. In §6, we then conduct a mixed quantitative and qualitative investigation of focal subjects, building on insights from Lambrecht & Polinsky (1997) and others regarding the properties associated with focal subjects.

5 A multivariate approach to Differential A Indexing

The preceding descriptive analysis of accessibility-related factors provides a reasonable approximation of the factors behind Differential A Indexing in Hewramî, but it requires validation against a statistically more rigorous approach that considers other factors identified in the literature (Ouhalla, 1993; Lambrecht & Polinsky 1997; Siewierska 2004b; among others). In what follows, we apply a decision tree model, which assesses the respective contributory effects of multiple predictor variables and provides intuitively interpretable visualizations.

Conditional inference trees (CITs) are models that are developed through the data, hence the structure of the model is not defined a priori. They make predictions on the outcomes based on dependent variables, in this case, “index” and “no index”. In other words, they test recursively how the presence or lack of verb indexing (dependent variable) is influenced by predictor variables. We used the `ctree()` function from the `party` package in R (Hothorn et al. 2006) to construct conditional inference trees. Conditional inference trees are better suited for handling some of the key limitations of traditional decision trees (e.g., classification and regression trees), such as variable selection bias and overfitting. Traditional decision trees tend to favour predictors with many possible splits, but conditional inference trees use permutation-based hypothesis testing to select variables (with p-values obtained from permutation tests), ensuring unbiased and more reliable decision-making. This separation of variable selection from split point determination leads to models that are not only more interpretable but also better suited for datasets with mixed variable types or missing values. As a result, they provide a more principled and robust framework for classification and regression tasks.

The dependent variable is presence versus absence of an A-index in a past transitive clause. We have already established that clauses with zero A are almost categorically indexed, so we restrict the sample to only those clauses which contain an overt subject constituent, as these are the ones which exhibit a relevant degree of variation regarding the dependent variable. Table 8 provides an overview of the variables. In §5.1, we present a brief discussion of each predictor variable and a summary of the relevant raw data, before presenting the results of the combined model in §5.2. The raw figures for the variable ‘PoS of A’ (Part of Speech of A) have already been presented in Table 6 above and are not repeated here.

5.1 Individual factors

5.1.1 *Constituent order*

Hewramî is OV generally, but exhibits some flexibility, which is largely pragmatically determined. The focus position in Hewramî is immediately preverbal and in pragmatically neutral transitive clauses is occupied by the O; we provide a detailed account of focus in Hewramî in §6 below. Given that Differential A Indexing is known to be sensitive to information structure, word order is taken here as a rough proxy for information structure in the model. Some clauses in our sample lack an overt O. There

Table 8. Variables

	Variables	Values
Dependent variable	Indexing	index, no index
Predictor variables	Constituent order	AV, OAV, AOV, VOA, VA
	Case marking of A	direct, oblique, na (for first and second persons), nv ('non-visible' for NPs ending in <i>ê</i> , <i>î</i> or followed by the additive particle = <i>îç</i>)
	Animacy of the A	human, non-human animate, inanimate
	Clause type	main declarative, interr(ogative), comp(lement), adv(erbial), rel(ative), cond(itional),
	PoS of A	noun, indf-pro, wh-pro, pro-1, pro-2, pro-3

Table 9. Indexing and word order constellation

Word order	Total clauses	A is indexed		A is not indexed	
		N	%	N	%
OAV	28	4	14	24	86
AV	183	74	40	109	60
AOV	174	156	90	18	10
OVA	10	9	90	1	10
VA	15	14	93	1	7

are five values attested for this feature: AV; OAV; AOV; VOA; VA. The raw figures for each are provided in Table 9.

According to the figures in Table 9, subject indexing is most likely in those word orders where O is in the immediately preverbal position (AOV, OVA), or the preverbal position is not occupied by A (VA). Indexing is least likely in any word order where A is in the immediately pre-verbal position (AV, AVO). This is already suggestive of a crucial role of focality of A as a factor inhibiting indexing. We take up the issue of word order and focus in §6.

5.1.2 Case

As discussed earlier, Hewramî has ergative alignment in the past tense. The expected ergative marking (via the oblique case) of the A in the ergative construction is, however, not always realized. There are three classes of subjects which lack the expected oblique case. First, it is systematically absent on first and second person pronouns. For these subjects, the value “na” (not applicable), is assigned in the annotation. Second, for feminine nouns ending in *ê*⁶ and A NPs followed by the additive particle =*îç* the case

⁶ The case suffix is also not visible following masculine nouns ending in *î*, but such nouns are not attested in the A-past function in our data.

Table 10. Indexing and flagging, distinguishing nouns and pronouns

	N clauses	Oblique		Direct	
		A indexed	A not indexed	A indexed	A not indexed
Nouns	235	109 (55%)	91 (45%)	33 (94%)	2 (6%)
Pronouns	46	15 (47%)	17 (53%)	14 (100%)	0 (0%)
Total	281	124 (53%)	108 (47%)	47 (96%)	2 (4%)

distinctions are not visible. For these subjects, the value “nv” (not visible), is assigned in the annotation. Third, the absence of overt A-marking may be triggered by largely discourse-pragmatic factors (see below). Hewramî thus exhibits not only Differential A Indexing but also Differential A Marking in the past tense (Mohammadirad 2024). Given that the presence of indexing is also (partially) conditioned by extra-linguistic factors, we expect complex interactions between differential case marking and differential indexing, which can be more meaningfully analysed in concert with the other predictor variables in the model, and are discussed in §6 and §7. The basic Hewramî data on flagging and indexing are provided in Table 10, which differentiates nouns and pronouns (note that the figures for pronouns contain both third person pronouns and indefinite pronouns). The first and second person pronouns as well as feminine nouns ending in *ê* and A NPs followed by the additive particle *=îç* are not counted since the case distinction is either not available or not visible, which means the total number of tokens included (281) is lower than the overall number of tokens with overt A (410).

Table 10 shows that while lack of oblique case marking strongly predicts the presence of indexing (96%), no prediction can be made for oblique case-marked As, which are approximately equally likely to be indexed or not.

5.1.3 Animacy

Recall that in accessibility-based accounts, humans are considered more salient than animate and inanimate referents (human > non-human animate > inanimate), and cross-linguistic studies have also suggested that agreement relations are sensitive to animacy (Ariel 1990). Therefore, we expect there to be more agreement with items higher in animacy. Table 11 summarizes the results for the link between the absence of indexing and the animacy of the A-past NPs. The figures represent the total number of arguments regardless of case marking. The figure for ‘Human’ includes human proper nouns and pronouns with human reference (speech act, third person, indefinite, and interrogative pronouns).

The difference between animates and inanimates barely reaches significance, but if we combine the two into a superordinate category “non-human”, as suggested, for instance, in Haig & Schnell (2016), there is a highly significant difference to the “human” category (p-value of 0.00012 on two-tailed Fisher’s exact test). This indicates that, when reduced

Table 11. Animacy of overt A-past arguments

Animacy	N clauses	A indexed		A not indexed	
		N	%	N	%
Human	365	241	66	124	34
Non-human animate	31	11	35	20	65
Inanimate	14	5	36	9	64

Table 12. A-past indexing across different clause types

Clause type	N clauses	A indexed		A not indexed	
		N	%	N	%
Subordinate clauses	43	24	56	19	44
Main clauses (including interrogatives)	367	233	63	134	37

to the distinction “human versus non-human”, animacy is a significant factor in determining the presence of indexing.

5.1.4 Clause type

Finally, the last predictor variable for Differential A Indexing is the clause type. According to Table 12, there is less A indexing in subordinate clauses than in main declarative clauses, but the difference is not statistically significant (Fisher’s exact test p -value equals 0.32). In §7 below, we take a finer-grained look at sub-types of subordinate clauses, which exhibit quite variable profiles.

5.2 Results of the decision tree analysis

Figure 3 illustrates the result of the combined model when all variables are considered for the presence of indexing (illustrated in dark shading) as opposed to its absence (illustrated in grey). The predictor variables are represented by nodes. Nodes, in turn, are split based on statistical significance and along the values of a predictor. The tree captures the hierarchical structure of the relationships among variables, and it incorporates interaction effects between them. According to this model, constituent order is the strongest factor affecting the presence of indexing. The model suggests an initial split between orders in which A is in the immediately preverbal focus position (AV, OAV), and where it is not (AOV, OVA, VA). For the first group, case marking (Node 5) is the next best predictor, while animacy (human vs. other, Node 7) becomes influential for oblique case-marked, “na” and “nv” A. For the second group, with A outside the

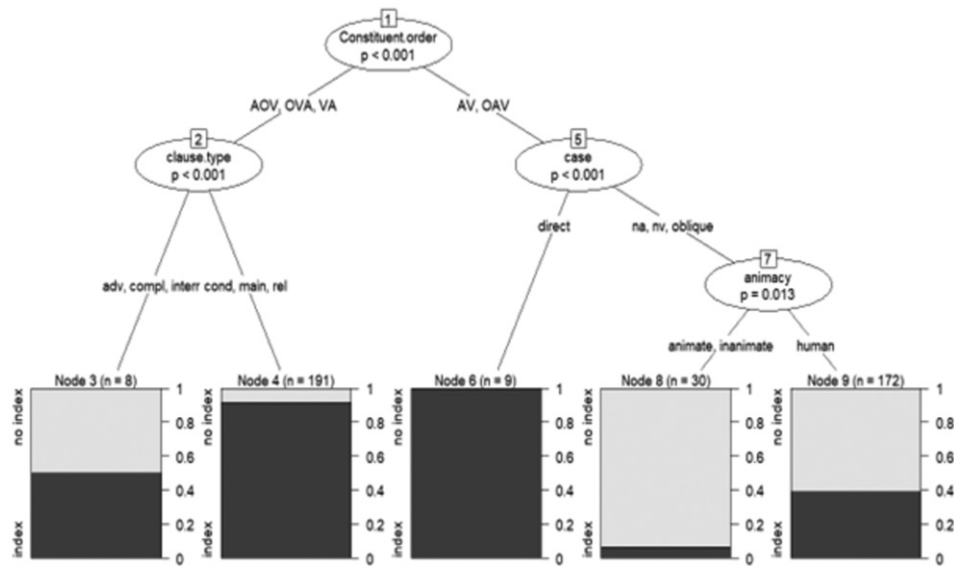


Figure 3: Predicting the presence of subject indexing

immediately preverbal focus position, indexing is generally present. There is also an effect of clause type (Node 2) as a fairly weak predictor of absence of indexing (see §7).

In short, the strongest predictor is whether the A is in the canonical focus position (immediately preverbal), thus AV, or OAV or not. In all other constituent orders, either the focus position is occupied by the O, or is empty and the A is in an after-thought position post-verbally (VA). The model thus identifies a strong effect of subject focus, which we explore in more detail in §6. Somewhat surprisingly, the predictor PoS, which differentiates pronouns and lexical NPs, does not figure as a significant variable in the model. It thus appears to have no particular effect whether the A is pronominal or lexical, but note that pronominal here includes indefinite pronouns, which may skew the results. The predictor Animacy only appears relevant under specific conditions; therefore, it calls for some additional explanation. First, it should be noted that the binary distinction implemented in the model, human vs. non-human (including inanimate and animate non-human), splits the data very unevenly: 11% non-human (N = 45 out of 410 clauses), 89% human (see Table 11).⁷ Of the total 45 non-human A, 71% (32) occur with the AV and OAV word order, suggesting that non-human subjects are overall more likely than chance to be associated with this word order (the overall frequency of this word order in the data is 51%). For these 32 tokens, indexing is overall strongly dispreferred (28 of 32 no index). Of the 32, 28 are also case-marked, and 93% of these are not indexed. In other words, an A that is non-human and has the AV and OAV order (focal word order) is

⁷ These proportions lie squarely within the range of values from cross-linguistic data on the frequency of non-human transitive subjects in spoken discourse (< 15%) cited in Haig et al. (2022: 164).

Table 13. Evaluation of the conditional inference tree model

Metric	Value
Prediction accuracy	79.7%
Balanced accuracy	78.6%
Sensitivity (recall for “index”)	83.8%
Specificity (recall for “no index”)	73.5%
Positive predictive value (precision for “index”)	82.7%
Negative predictive value (precision for “no index”)	75.0%
Prevalence of “index”	60.2%
Detection rate (correct predictions of “index”)	50.4%
Detection prevalence (predicted “index”)	61.0%
McNemar’s test p-value	1.0

likely to be both oblique-marked and not indexed. But the CIT’s prediction of “no index” is more uncertain for humans. More importantly, what the CIT shows is that the significant differences observed for animacy in Table 11 do not carry over in the multivariate analysis. Once constituent order is taken into account, animacy becomes basically non-influential to predict differential indexing.

The conditional inference tree was evaluated for its ability to predict the presence or absence of indexing. The prediction accuracy is 79.7%, meaning 8 out of 10 predictions are correct. This is significantly higher than the baseline for subject indexing in the corpus, which is 60.2%. A 95% confidence interval for the model is between 71.5% and 86.4%. A one-sided binomial test comparing the model’s accuracy (79.7%) to the baseline accuracy of 60.2% (the prevalence of the majority class, *index*) gives a p-value smaller than 0.0001, showing that the model’s performance is significantly better than random estimates.

Other metrics are available in Table 13. The positive predictive value (precision) for “index” was high at 82.7%. In comparison, the negative predictive value for “no index” was lower at 75.0%, suggesting that the model has more confidence in predicting “index” than “no index”. The prevalence of the index class was 60.2%. The model predicted “index” 61.0% of the time (detection prevalence), detecting it correctly in 50.4% of all cases (detection rate). The McNemar’s test returns a p-value of 1.0. This suggests that the model’s errors do not show significant asymmetry between the two classes – the model does not consistently misclassify one class more than the other.

6 Focal subjects

The model suggests that Differential A Indexing in Hewrami is strongly determined by information structure. In particular, there is an association between subjects in the typical focus position (immediately pre-verbal) and a lack of indexing. First, we consider clauses where A is an interrogative pronoun. Although there are differing approaches to the definition (and indeed validity) of ‘focus’ as a cross-linguistically relevant category (see

Matić & Wedgewood 2013), it is widely agreed that interrogative elements are associated with focus (but see Erteschik-Shir 1986 for a different perspective). Furthermore, subject interrogatives have been suggested to be those most likely to be associated with non-canonical syntax (Skopeteas & Fanselow 2010). There are seven tokens of subject WH-questions in the data; five of them lack indexing, for example (11) and (12).

- (11) *î* *zeř-e* *çerme=m=e* *kê*
 DEM.PROX money-EZ.CMPD white=1SG=DEIC who
berde=n *eçêge*
 take.PST.PTCP.M=COP.3SG.M:O here
 ‘Who has taken my silver money [that is now] here?’ [PK.29]

- (12) *wat=şa* *daxom* *kê* *etik* *kerdê=b-o*
 say.PST=3PL:A Q.PTCL who disgrace do.PST.PTCP.F=be.PRS.3SG:O
 ‘They said: ‘We wonder who might have disgraced her?’ [ED.195]

Examples (13) and (14) are consecutive utterances from the same text. Each consists of two contrasted clauses, and in each, the first clause includes an index while the second lacks one.

- (13) *kê* *řenc=iş* *da=w* *kê* *berd*
 who toil=3SG:A give.PST=and who take.PST
 ‘(Look) who toiled and who took (the credit)!’ [YX.15]

- (14) *min* *řenc=im* *da=w* *tò* *berd*
 1SG toil=1SG:A give.PST=and 2SG take.PST
 ‘I toiled, and you took (the credit)!’ [YX.16]

Looking at the second clauses in each, both involve a break in subject continuity and an element of contrast with the subject of the immediately preceding clause (‘who was it who took...’ and ‘it was you who took...’). The lack of indexing with the second conjunct both in (13) and (14) can be explained by the unexpectedness of the contrasted subject. A focus interpretation is further underscored by the immediately pre-verbal position of the subject. A very similar use of an emphatic focus is found in (15), again with immediately pre-verbal position of the pronoun and with the sense of ‘it was you who...’.

- (15) *m-aç-o* *wila* *to* *kuştê=nê*
 IND-say.PRS-3SG:A indeed 2SG kill.PST.PTCP.PL=3PL:O
 ‘He said, ‘Yes indeed, YOU have killed them. [You are right].’ [ME.157]

The absence of indexing for A is also seen when A is in a polar question. In (16), the focused 2SG A argument in the polar question is not indexed.

- (16) *m-aç-o* *milk* *ehmedî* *î* *dijmen-ê-m=e* *kê*
 IND-say.PRS-3SG:A PN PN DEM.PROX enemy-PL.DIR-1SG=DEM who
kuştê=nê? *tò* *kuştê=nê?*
 kill.PST.PTCP.PL=COP.3PL:O 2SG kill.PST.PTCP.PL=COP.3PL:O
 ‘He (the king) said, ‘Oh Milk Ahmad! Who has killed my enemies? Have YOU killed them?’ [ME.150]

These examples suggest that a focus construction will inhibit indexing, even with Speech Act Participant subjects, although the latter are overall highly likely to have indexing (79%). Another kind of focus is contrastive focus, where a particular referent is highlighted from set of potential candidates, often with a sense of counter-expectation or surprisal. In (17), the speaker's assertion counters the addressee's assumption that the speaker's son has been killed by the Rozgaris.

- (17) *kuř-û min řozgarî-a ne-kuşte=n*
 son-EZ.GEN 1SG PN-PL.OBL NEG-kill.PST.PTCP.M=COP.3SG.M:O
xwa-γ kuşte=n
 God-M.SG.OBL kill.PST.PTCP.M=COP.3SG.M:O
 'The Rozgaris did not kill my son; God killed him.' [HM.08]

Notice that the first clause in (17) contains a fronted topical object, 'my son', bringing the subject into the preverbal focus position and inhibiting the expected index. This is indeed the hallmark of the majority of non-indexed subjects, as in examples (18)–(20).

- (18) *kuř-û min xuda-γ kuşte=n*
 son-EZ.GEN 1SG God-M.SG.OBL kill.PST.PTCP.M=COP.3SG.M:O
 'God killed my son.' [HM.95]
- (19) *werês(e)-ekê min wardê=ne*
 rope-DEF.F.DIR 1SG eat.PST.PTCP.F=COP.3SG.F:O
 '[The lion said], 'I ate the rope!'' [ÇH.85]
- (20) *zeř-e çerme=yç adê ne-warde=n*
 coin-EZ.CMPD white=ADD 3SG.F.OBL NEG-eat.PST.PTCP.M=COP.3SG.F:O
 'It (the cat) has not eaten the silver coin(s).' [PK.23]

When the subject is already established as a discourse topic, it may optionally occur as a post-verbal afterthought constituent, in which case it is outside the focus domain, and the clause includes a subject index, as in (21) and (22) – see also the broader context of these examples.

- (21) They said, "Pharaoh, **a box** has arrived floating on the river."
asaw-eke=ş bîna=n qot(e)-ekey
 mill-DEF=3SG:A block.PST.PTCP.M=COP.3SG.M:O box-DEF.M.SG.OBL
 'The box has blocked the (water) mill.' [MF.75]
- (22) **Rosam** went and put his finger inside the hole of the millstone [which was blocking the prison door]...
bêjen=iş gêrt tabêl řosem-î
 PN=3SG:A grab.PST delivery PN-M.SG.OBL
 '[Then], Rosam got hold of Bezhan.' [BM. 115]

These examples illustrate that lack of indexing is associated with a particular information structural configuration, which we provisionally refer to as focal subjects. Focal subjects

have been linked to Differential A Indexing (i.e., a lack of expected subject agreement, see Siewierska 2004b: 159–162 and Just 2024: 304 for examples). Furthermore, a number of researchers have identified focal subjects as a locus of non-canonical syntax (see Skopeteas & Fanselow 2010 for experimental evidence). Lambrecht & Polinsky (1997) enumerate the properties associated with focal subjects cross-linguistically, including (i) prosodic prominence; (ii) specific linear position; (iii) non-nominative case marking; and (iv) lack of grammatical agreement.

In Hewramî, a specific linear position (immediately pre-verbal) and lack of indexing are also exhibited by focal subjects. The prosodic characteristics of focal arguments in Hewramî have been analysed in Mohammadirad (2026), but for reasons of brevity they are not taken up here. With regard to case marking, oblique marking (Lambrecht & Polinsky’s 1997 “non-nominative”) of A is in fact overall more likely for focal (immediately pre-verbal) A arguments, and this effect is amplified for non-human arguments (see below).

Focality alone is of course not the whole story. But overall, our results align with cross-linguistic findings on focus and on Differential A Indexing. More generally, we would state that indexing is less likely for subjects that lack the properties of an immediate discourse topic at a given point in discourse. An A having the same referent as the subject of the immediately preceding clause is most likely to be expressed through zero anaphora in Hewramî, and as we have seen, that is precisely the environment where a pronominal index is almost categorically present. Subjects that diverge from this in varying ways, because, for example, they represent unexpected or contrastive information, or a request for information, or are part of athetic clause (e.g., (24)), are the ones most likely to lack indexing. This is also true for the few tokens of indefinite subject pronouns in the sample, as in (23) and (24).

- (23) *kes-î be noŋ-ê pene ne-zana=n*
 no_one-M.SG.OBL to kind-INDF to NEG-know.PST.PTCP.M=COP.3SG.M:O
 ‘No one really appreciated him.’ [ZP.24]

- (24) *jo-y bawŋi kerd-a pore*
 one-M.SG.OBL hug do.PST-1SG:O ADP
 ‘Someone hugged me.’ [KT.97]

In §8, we evaluate topic shift constructions, but first we briefly consider animacy and clause type.

7 Other factors: Animacy and clause type

The analysis in the decision tree model discussed in §5 shows that animacy and clause type are relevant for predicting the differential indexing of the A argument, though only under certain conditions. Considering animacy, the primary division is between human and non-human, while animate vs. inanimate is not significant (see Table 11). Also note that according to the CIT model, the prediction of “no index” is more uncertain for humans than for non-humans. From this it follows that the most likely kind of

non-indexed subject in Hewramî is immediately pre-verbal (focal), non-human, and oblique case-marked (or lacking a case distinction). Examples (25)–(27) are illustrative.

- (25) *mar-êw-î* *gest-a*
 snake-INDF-M.SG.OBL bite.PST-1SG:O
 ‘A snake bit me.’ [MP.09]
- (26) *îse* *caw(e)-ê* *kerdê=ne* *çêr=o*
 now road-F.SG.OBL do.PST.PTCP.F=COP.3SG.F:O under=POST
 ‘Now, the road has covered it (the stone).’ [ZP.53]
- (27) *pejare-y* *berd-o=me*
 grief-M.SG.OBL take.PST-3SG:O=COMPL
 ‘The grief overtook him.’

Our data suggest a subtle interaction between oblique marking and animacy (or humanness), which can be related to the findings for the Hewramî dialect of Luhon (Mackenzie 1966: 51). In that dialect, apparently only inanimate As are oblique-marked and lack indexing.⁸ In our data from the Tekht variety, however, oblique marking is still available for a human A, as in (28)–(31), and an animate A, as in (25). The difference between the Luhon variety and our Tekht variety could be interpreted in terms of a diachronic process in which the oblique marking of the A gradually recedes along a scale of animacy from highest to lowest. The Tekht variety retains an earlier stage of oblique marking on a wider range of A, including human and non-human, but already with a statistical preference for non-human (see above). In Luhon, this process has proceeded further, so that only inanimates are oblique-marked. This line of reasoning would be in line with Silverstein-type explanations for animacy-mediated split ergativity, according to which ergative case marking is most likely for an A that is lowest in animacy and presumably therefore retained there longest. The general lack of an oblique case on SAP pronouns in Hewramî could also be related to this: the items highest in animacy lose oblique marking first.

Regarding the interaction with indexing, in Luhon it is the norm for past transitive constructions but is absent in the (rare) presence of an oblique-marked A. For the variety of Tekht under investigation here, indexing may co-occur with an oblique-marked and human A, as in (28)–(31), which would be near impossible in Luhon.

- (28) *jînê=ç* *wat= îş*
 woman.F.SG.OBL=ADD say.PST=3SG:A
 ‘The woman said.’ [HS. 77]

⁸ Mackenzie (1966: 51) briefly distinguishes two kinds of past transitive constructions in the Luhon variety: Type 1 is relatively marginal, involves an “impersonal” (approximately “inanimate”) NP in the A role (e.g., “heat”, “rain”, “hunger”), which may take the oblique case, is not indexed, and occupies the immediately pre-verbal position. Type 2 is “by far the more common” construction and involves a Direct-marked A, which is always accompanied by an index and is clause-initial. We have not yet been able to verify the respective frequencies of these types with corpus data.

- (29) *padša-y desur=iş da=n be min*
 king-M.SG.OBL order.M=3SG:A give.PST.PTCP.M=COP.3SG:O to 1SG
 ‘The king has given me an order.’ [JP. 209]
- (30) *kuř-ekē-y meme-w bize-(e)kê=ş ward-o*
 boy-DEF-OBL.M.SG udder.M-EZ.GEN goat-DEF.F=3SG:A eat.PST-3SG:O
 ‘The boy fed from the goat’s udder.’ [KŞ. 031]
- (31) *xanmê kerd=iş=we cūwab*
 lady.F.SG.OBL do.PST=3SG:A=COMPL reply
 ‘The lady replied.’ [SH.148]

What this suggests is that the diachronic trajectory of indexing a past A, and oblique marking of a past A, are diametrically opposed in Hewramî – and possibly other Iranian languages: indexing is spreading to become the norm for all past transitive constructions, while oblique marking of the A is disappearing. Luhon would appear to be further in both developments, having almost entirely lost oblique marking, while indexing is the norm for the commonest type of past transitive construction (see footnote 5). Tekht, on the other hand, retains oblique with a larger class of A (including animates and humans), with indexing already nearly as widespread as in the Luhon system. The endpoint of this trajectory is found in neighbouring Central Kurdish, where oblique marking of the A has completely disappeared, and indexing is fully obligatory, regardless of animacy or information structure.

As for the factor clause type, it has been reported that in some languages differential A indexing is tied to negative clauses, interrogatives, complement clauses and relative clauses (Ouhalla 1993). The decision tree in Figure 3 does identify clause type as a significant factor, but only for clauses with non-focal subjects, and not along the boundaries of main versus subordinate. Rather, it seems that conditional, relative, and main clauses cluster together against interrogative, adverbial, and complement clauses. Table 14 breaks down the raw figures for different kinds of main (including interrogative and declarative) and subordinate clauses (the remaining clauses).

It is not obvious how these figures would relate to the information structural account developed above, because these clause types are not consistently associated with specific information-structural characteristics. For example, relative clauses will vary according to the role of the relativized constituent. In (32)–(34) we find object relativization with the object omitted (but reflected through agreement morphology on the verb), leaving the A of the relative clause in the immediately pre-verbal position. The lack of indexing in these clauses would then fall out straightforwardly from the word order effects already alluded to.

- (32) *î gîr=e çêş bî-Ø min ward-Ø*
 DEM.PROX hook.M=DEM what be.PST-3SG.M:S 1SG eat.PST-3SG:O
 ‘What is this situation that I am caught in?’ [Lit. What is this hook that I ate] [HB.23]
- (33) *kuř-û şuwane-î heke to kuşt-e=û*
 son.M-EZ.GEN shepherd.M-OBL.M REL 2SG kill.PST-COND.3SG:O=and
 ‘The shepherd’s son who you [wanted to be] killed’ [KŞ.51]

Table 14. Indexation of overt A-past arguments according to clause type

Type of subordinate clause	N clauses	A indexed		A not indexed	
		N	%	N	%
Complement clauses	9	1	11	8	89
Adverbial clauses	7	2	29	5	71
Interrogative clauses	8	3	37	5	63
Relative clauses	16	11	69	5	31
Conditional clauses	11	10	91	1	9
Main declarative clauses	359	230	64	129	36

- (34) *ane key b-ê dizî şime kerqê=n*
 DEM.DIST how be.PRS-AUG.3SG:S theft 2PL do.PST.PTCP.M=COP.3SG:O
 'How could that be considered a theft that you have committed?' [ED.107]

With subject relativization and the object in the immediately pre-verbal position, indexing is found (35).

- (35) *aney sere-(e)ke=s wardê=b-ê*
 DEM.3SG.M.OBL head-DEF.M.SG.DIR=3SG:A eat.PST.PTCP.M=be.PRS-AUG.3SG:O
luma řa-w hat-î=re
 go.PST.3SG:S road.F-EZ.GEN fortune-M.SG.OBL=POVB
 'The one who had eaten (the dove's head) took (lit. went to) the way of good fortune.'

In complement clauses, indexing is generally lacking (eight of nine examples), as in (36). This suggests that complement clauses are generally contexts that inhibit indexing, regardless of any other features.

- (36) *pêse waç-î xma-y ket-ê pey*
 as_if say.SBJV.PRS-2SG:A God-OBL.M bed-INDF to
kîyase=n
 send.PST.PTCP.M=COP.3SG.M:R
 '[He had levitated this much from the ground], as if [you said] God had sent him a bed.' [JP.69]

This property of subordinate clauses could be a reflection of their tendency to be more resistant to syntactic change, as suggested by Bybee (2001). She relates this to the fact that they are processed in large chunks, making their constituents less independent and less likely to change (see Givón 1983, 1990 for a somewhat different account in terms of discourse cohesion). However, due to the low absolute numbers of complement clauses, interrogatives, conditionals, and adverbial clauses ($N < 20$), we refrain from speculative interpretations of the variation in these categories.

To sum up, we note an effect of animacy on indexing, such that a non-human A is less likely to be indexed than a human A. With oblique case marking, there is an approximately reverse tendency, such that a high animacy A is least likely to be

oblique-marked. Overall, then, the most likely kind of non-indexed subject in Hewramî is immediately pre-verbal, non-human, and oblique case-marked (or lacking a case distinction). A comparison with the Luhon dialect of Hewramî (MacKenzie 1966), where only inanimate A is case-marked, and only these lack indexing, leads us to hypothesize that indexing of the A is spreading, while oblique marking of the A is retracting, and both processes are further advanced in the Luhon dialect.

8 Topic shift and additive-marked subjects

Recall from the discussion in §2 that according to Givón (1976, 2017), the grammaticalization of subject agreement begins with the de-pragmatization of a “topic shift” construction, originally used in contexts where a previously mentioned, but no longer topical, entity is recalled into the centre of the discourse flow, approximately corresponding to an ‘as for’ phrase in English. Over time, this construction is reanalysed as a pragmatically neutral one, and the co-occurrence of overt subject NP (the original fronted topic) and a resumptive pronoun grammaticalizes to a subject-agreement construction. In this section, we present the results of a quantitative study of indexing in topic-shift constructions in the Hewramî data. The hypothesis to be tested is that if topic shift constructions are the first to develop indexing diachronically, then they should be among the contexts with the highest frequency of indexing in contemporary language.

In the Hewramî corpus, such constructions are overall rather infrequent. The closest candidates are clauses in which the subject is marked by the so-called additive particle =îç, which, among other things, marks the topic in a topic-shift construction (see Forker 2016 on the typology of additive particles). Example (37b) is a typical instance of a topic-shifted subject marked by the additive particle; there is no subject index.

- (37) a. *xeber=şa* *da-Ø* *patşa-y*
 news=3PL:A give.PST-3SG.M:O king-M.SG.OBL
 ‘They passed the news to the king.’ [DB.310]
- b. *patşa-y=ç* *xutke* *kerd-Ø*
 king-M.SG.OBL=ADD invitation do.PST-3SG.M:O
 ‘The king invited him.’ [DB.311]

Table 15 shows the overall frequency of indexing in all clauses with a left-dislocated topic-shift construction with an additive particle, such as (37b), in our corpus.

The frequency of indexing in topic shift constructions is 59%, slightly below the overall baseline level for indexing in clauses with an overt subject (63%). There is thus no evidence that topic shift constructions are significantly more likely to trigger indexing than any other kind of clause with overt subject NP. If topic shift constructions were the first constructions in which indexing became established, we would expect higher frequencies in our data.

Topic shift constructions with the additive particle =îç only occur in low absolute frequency in our data (less than 3% of the total 2005 transitive clauses). In order to test for an overall effect of topic continuity, we applied a broader definition to identify clauses

Table 15. Indexing and topic-shift constructions marked by the additive =iç

	Total	A indexed		A not indexed	
		N	%	N	%
Clauses with topic shift =iç	44	26	59	18	41

with discontinuous topics, operationalized as: “any (past transitive) clause containing an overt subject constituent that is not co-referent with the subject of the immediately preceding clause.” On this definition, the presence of the additive particle is no longer constitutive. An example of this kind of discontinuous topic construction is (38b), where the subject *Siyawahsh* is different to the 3PL subject of the preceding clauses. Note, however, that *Siyawahsh* is mentioned in the preceding sentence, but not as the subject.

- (38) a. *dêw-a berd=şa siyamehş-i bûş-a*
 ogre-PL.OBL take.PST=3PL:A PN-M.SG.OBL kill.PRS.SBJV-3PL:A
sere=ş bîr-a
 head=3SG:POS cut.PRS.SBJV-3PL:A
 ‘The ogres took Siyawahsh [to a desert] to kill him, to behead him.’ [SK.65]
- b. *siyamehş-i fermawa=ş*
 PN-M.SG.OBL say.PST=3SG:A
 ‘Siyawahsh said [...]’ [SK.66]

Clauses that satisfy this definition actually make up almost three quarters of the total clauses with overt subjects. This is not unexpected if we recall that the normal pattern for topic continuity is for zero subjects. In other words, the presence of any kind of overt subject is in itself a reasonably reliable cue for topic discontinuity. Table 16 shows the overall rates of indexing in clauses with discontinuous topics, differentiated according to the word order of the discontinuous topic.

Table 16 considers only left-dislocated (or more generally preverbal) discontinuous topic subjects. A small number of clauses with discontinuous topics has VA order (N = 16), as in (39).

- (39) *hênas(e)-êw-i serd=iş kêşt patşa-y*
 sigh.M-INDF-EZ.ATTR cold=3SG:A pull.PST king-M.SG.OBL
 ‘The king breathed a heavy sigh.’ [DB.325]

We have already noted in §5 that clauses with VA order overwhelmingly prefer indexing. That could be explained by assuming that the post-verbal subject is outside the core of the clause, leaving the clause without an overt subject, thus almost categorically required to include an index. At any rate, this word order cannot be the point of departure for the current past transitive construction, which is overwhelmingly AV, so we disregard right-dislocated topics in the analysis here.

Table 16. Discontinuous topics and the indexation of A-past arguments

Clauses with discontinuous topics not marked with =i _{cl}	Total	A indexed		A not indexed	
		N	%	N	%
Word order: A ... V	269	152	57	117	43

Overall, we find no evidence for a higher frequency of indexing in connection with discontinuous topics, either left-dislocated with the additive clitic, or simply expressing a different subject to the preceding clause: Frequencies of indexing hover slightly below the baseline rate of 63% for all clauses with overt subjects. From this we conclude that the grammaticalization of subject agreement was not connected to a pragmatically-marked construction used to indicate a shift from the current topic. Rather, we believe that subject indexing first became established in clauses with topic continuity and no overt subject constituent – in fact the most frequent kind of transitive clause in discourse. The index essentially “filled a gap” in these clauses, which otherwise lacked any overt representation of the person and number features of the subject. Essentially, indexing in the past transitive construction was restored to be in line with the rest of the grammar, where subject indexing is obligatory. Topic shift constructions such as (37b) would have been available as signals of disruption to topic continuity throughout the history of the language and are still available today. But we see no reason to assume that they played any particular role in kickstarting this development. Based on the available data, we can state with some confidence that indexing is not significantly associated with a pragmatically marked topic shift construction, and such constructions are therefore an unlikely point of departure for the development of grammatical agreement.

9 Conclusions

Iranian languages underwent a wholesale loss of subject agreement in past transitive constructions in the Middle Iranian period. The majority trend for contemporary West Iranian languages has been to restore fully obligatory subject agreement through various strategies (see Figure 1). A small number of languages, including Hewramî of Tekht, nevertheless leave a subset of subjects non-indexed. Our detailed exploration of the factors conditioning the presence of an index reveals much that confirms conventional wisdom regarding the spread of agreement, in particular the role of accessibility-enhancing factors such as person (first and second person subjects are more likely to be indexed than third person), humanness, and more generally topicality in the sense of topic continuity. The main factor that inhibits indexing can be approximately identified as focality of the subject, or more generally, subjects that fail to comply with the information-structural default for subjects of continuing the preceding topic. These atypical subjects include interrogatives, contrastive foci, and subjects ofthetic sentences. Overall, we find that indexing is most likely to be absent with subjects that are focal, non-human, and oblique case-marked. Our findings regarding the lack of indexing on

focal subjects aligns with research that identifies focal subjects as a locus of non-canonical syntax (Skopeteas & Fanselow 2010) and are likewise consistent with existing research on systems of differential subject indexing.

The most striking finding in our data is the almost categorical presence of indexing in clauses that otherwise lack an overt subject NP – precisely the contexts characterized by topic continuity. Our results suggest that the clitic pronouns are initially deployed in those contexts where an anaphoric pronoun would *not* be expected – clauses with highly predictable, continuous topic subjects, where pro-drop is the norm. The process appears to be driven by a general drift towards a system of realizing at least one overt marker of person and number features of the subject in a clause. This is essentially in line with Berdicevski's et al. (2020) "trade-off" approach to indexing, which claims that indexing develops to maintain an efficient distribution of information approximately equating to one overt signal of subject identity per clause. Languages with obligatory agreement of course violate this ideal by involving more than one such signal in clauses containing both an overt subject NP and an index elsewhere in the clause. But such seemingly redundant over-marking is relatively infrequent in languages with high rates of pro-drop, such as Hewramî, and are apparently tolerated. What is not tolerated, in Hewramî at least, are clauses lacking any overt expression of subject person – *horror vacui*.

Even if we are prepared to accept that such a principle holds some degree of explanatory power for the developments in Hewramî, and for Iranian languages generally, it is clearly not a universal design feature of human language. As has been pointed out many times, languages of Mainland Southeast Asia (among others) maintain systems lacking any form of subject indexing over centuries or even millennia, and clauses lacking overt indication of subject person are rife in natural discourse. Thompson (1965: 293) describes the system in spoken Vietnamese, where the expression of relative social status takes precedence over the expression of person, as follows:

[...] the category of person is nearly an optional one in Vietnamese: little in the structure of the language itself forces the speaker to indicate specifically whether he refers to himself, to his listener or to another person. On the other hand, the status of the various persons involved must nearly always be clearly stated.

The tendency to prefer at least one overt expression of the category "person of subject" thus cannot be universal, as speakers of Vietnamese evidently can dispense with it. Rather, it is specific to certain areal and genetically defined language groups. In the case of Iranian, subject agreement has been a feature of the languages for over 2,000 years, and crucially, it has remained intact in most syntactic contexts – just not with past transitive clauses. Under these conditions, one might surmise that speakers are primed to expect information on the category of person for each clause, and structures lacking it are brought into synch with the general profile of the language. This would match with Bickel's (2003) notion of syntactic priming, "a well-established phenomenon whereby the processing of a particular structure facilitates processing of a subsequent utterance with the same or a similar structure [...]." In the case of past transitive clauses in Hewramî, "similar structures" would be, for example, transitive clauses in the present

tense, which still have rich paradigms of fully grammaticalized subject agreement, or any intransitive clause in the language, which likewise includes subject agreement. Taken together, these latter two clause types are far more frequent than the past transitive constructions under investigation here, which lends further credence to an account in terms of unifying the morphosyntax of past transitives with other clause types in the language.

These considerations make no appeal to the presence of a pragmatically-marked construction type serving as model for the overall development, and we have found no evidence that would support such an assumption. Constructions for signalling topic disruption and the establishment of a new topic were available in Middle Iranian times (Jügel & Samvelian 2020), and we have documented their continuing presence today. But we consider the presence of such constructions to be simply irrelevant in the overall development. Rather, it seems to have proceeded through an initial increase of frequency of pronominal clitics in clauses otherwise lacking an overt subject NP. Once established there, it spread to clauses with overt subject NPs, progressing approximately along a scale of more to least accessible (Table 6 and Figure 2). The least likely subjects to be indexed are non-human, oblique-marked, and in focus position.

Abbreviations

A	transitive subject	OBL	oblique
ADD	additive	PFV	perfective
ADP	adposition	PL	plural
ATTR	attributive	PN	proper noun
AUG	augment	POS	possessor
COMPL	completive	POST	postposition
COP	copula	POVB	post-verb
DEF	definite	PROX	proximate
DEIC	deictic	PRS	present
DEM	demonstrative	PST	past
DIM	diminutive	PTCL	particle
DIR	direct	PTCP	participle
DISC.PTCL	discourse particle	PUNCT	punctual
DIST	distal	PVB	preverb
EP	epenthesis	Q.PTCL	question particle
EZ	ezafe linker	REFLX	reflexive
EZ.CMPD	ezafe compound	REL	relativizer
EZ.GEN	genitive ezafe	S	intransitive subject
F	feminine	SBJV	subjunctive
GEN	genitive	SG	singular
IND	indicative	TOP	topic
INDF	indefinite	V	verb
INF	infinitive	VOC	vocative
M	masculine	1	first person
NEG	negative	2	second person
O	transitive object	3	third person

Ezafe is a head-linking strategy in the structure of the NP. The linker has different forms according to the category of the modifier.

Data Availability

The annotated corpus data used in this paper is publicly available in a CSV file in the Zenodo repository <https://doi.org/10.5281/zenodo.16936551> (Mohammadirad & Haig 2025).

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Author Contribution Statement

Masoud Mohammadirad recorded and transcribed the primary data, wrote the two initial drafts of the manuscript, compiled and tagged the dataset, and prepared it for deposit on Zenodo.

Geoffrey Haig co-wrote with MM subsequent revisions of the manuscript and contributed to the theoretical framing and the interpretation of the data.

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